DataGrep: AI-Powered Data Pipelines for Everyone

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Problem Statement

Over 80% of business professionals lack SQL or programming skills, yet organizations increasingly depend on data-driven decisions. When a marketing manager needs last quarter's revenue breakdown, they submit a request to an already-overloaded data team — and wait days or weeks for a response.

This bottleneck costs organizations time and agility. Simple analytics requests pile up alongside complex projects, creating backlogs that delay decisions across every department.

Solution

DataGrep lets any team member ask a plain-language question in Slack — like "What was our revenue growth last quarter?" — and receive an accurate answer with the right visualization in seconds. Behind the scenes, the platform interprets the question, constructs the necessary SQL queries, executes them in a secure sandbox, and delivers results directly in Slack. **No SQL knowledge required, no tickets, no waiting**.

Code

Demo

Try it out yourself

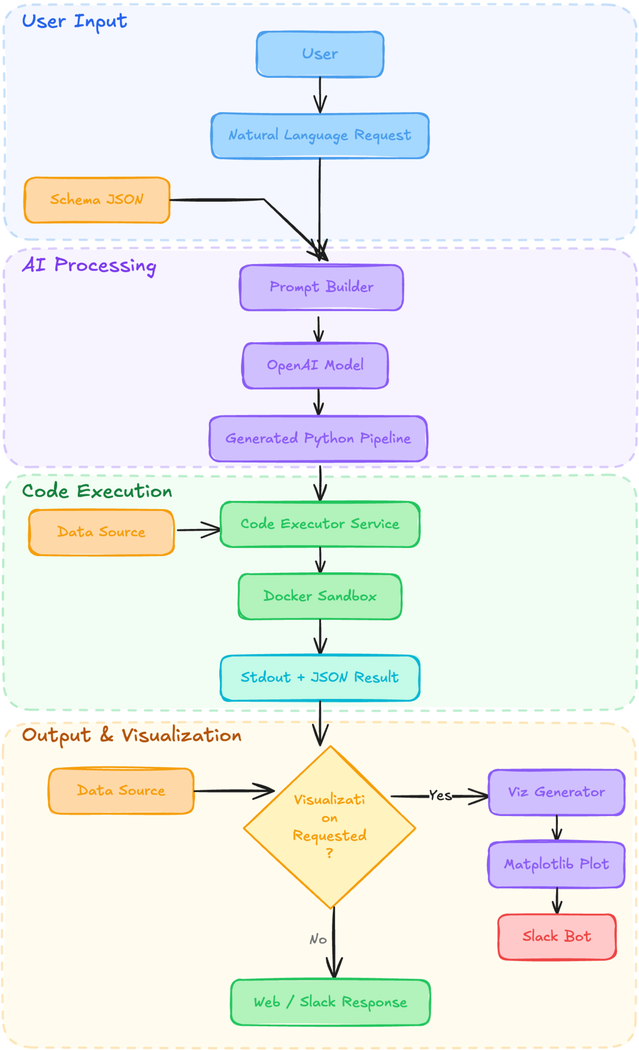
Why this approach

- **OpenAI Models** — Strong natural-language-to-SQL accuracy with flexible prompt engineering.
- **Docker Sandboxing** — Generated code runs in isolated containers, preventing untrusted queries from affecting production systems.
- **Slack Integration** — Meets users where they already work
- **Schema-Aware Prompting** — Ingests database schemas so the LLM generates accurate multi-table and source joins.

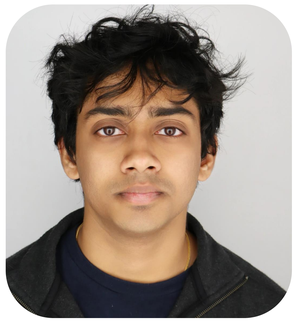
Successes and Challenges

DataGrep successfully handles complex, multi-table SQL queries triggered by conversational Slack messages, , correctly joining 3+ tables and producing accurate visualizations in testing.

The core results are consistent, but the generated SQL, code structure, and visualizations can vary between runs. Our next steps would be to standardize output frameworks, and validation layers that ensure reliable results.



Demo Flow



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